

# An Envelope-Fit Metric for Charge-State and Monoisotope Selection

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## Abstract

In untargeted MS1 feature detection, deciding which *charge state* and monoisotopic mass best explain an observed isotopic envelope is a model-selection problem. The current detector scores a hypothesis with an un-normalized weighted sum of comb-weighted, retention-time-smoothed intensities. That statistic rewards matched peaks but is blind to two failure modes: predicted isotope teeth that are *missing*, and observed peaks the hypothesis leaves *unexplained*. As a consequence it systematically mislabels a real charge-2 envelope (isotopes 0.5 Th apart) as charge-1 (isotopes 1.0 Th apart), because the  $z=1$  comb lands on every other peak and the raw sum does not care about the peaks in between. We propose a single unified score: the *cosine similarity* between the theoretical average envelope and the observed intensities, evaluated over the *union* of predicted-isotope positions and in-window observed peaks. We show that this one quantity simultaneously measures goodness of fit, the fraction of observed signal explained, and predicted-peak completeness — and that its square is exactly the fraction of observed signal energy captured by the best-scaled envelope. On a concrete peptide (EFLDVIK, true charge 2) the metric cleanly ranks the true charge above both the harmonic  $z=1$  and the over-split  $z=3$  hypotheses, where no single sub-metric does.

## 1 Motivation

### 1.1 The RawSum statistic and why it fails

For a charge/monoisotope hypothesis the detector currently accumulates

$$\text{RawSum} = \sum_k \sum_s w_k g_s I_{k,s}, \quad (1)$$

where  $k$  indexes the predicted isotope teeth of the hypothesis,  $s$  indexes scans across the retention-time (RT) apex,  $w_k$  is the average comb weight of tooth  $k$ ,  $g_s$  is an RT-Gaussian weight centered on the apex, and  $I_{k,s}$  is the observed intensity found at the  $m/z$  of tooth  $k$  in scan  $s$ .

Equation (1) is a *matched-filter response*. It grows with every predicted tooth that happens to find intensity, and it is never decremented. Two things it cannot see:

- (a) **Missing predicted teeth.** If a tooth the model requires has no intensity, that tooth simply contributes 0 to the sum. A hypothesis that explains half of its own predicted envelope is not penalized relative to one that explains all of it, as long as the absolute matched intensity is comparable.
- (b) **Unexplained observed peaks.** Any observed peak that does *not* sit on a predicted tooth is invisible to Eq. (1) entirely. A hypothesis is free to ignore half the signal in the window at no cost.

Because the score is neither normalized for completeness nor penalized for the signal it ignores, it does not distinguish a correct charge from a *harmonic* charge.

## 1.2 The harmonic example

Consider a genuine charge-2 peptide. Its isotope teeth are spaced  $1.00\text{-Da}/2 \approx 0.5\text{ Th}$  apart. A charge-1 hypothesis predicts teeth  $1.0\text{ Th}$  apart — i.e. on *every other* observed peak. Schematically, over eight observed peaks  $P_0, \dots, P_7$  spaced  $0.5\text{ Th}$ :

|                         |    |    |    |    |    |    |    |    |                                |
|-------------------------|----|----|----|----|----|----|----|----|--------------------------------|
| observed (0.5 Th grid): | P0 | P1 | P2 | P3 | P4 | P5 | P6 | P7 |                                |
| $z=2$ comb (true):      | ^  | ^  | ^  | ^  | ^  | ^  | ^  | ^  | (hits every peak)              |
| $z=1$ comb (harmonic):  | ^  |    | ^  |    | ^  |    | ^  |    | (hits every <i>other</i> peak) |

The  $z=1$  comb finds real intensity at  $P_0, P_2, P_4, P_6$  and produces a large **RawSum**. The intervening peaks  $P_1, P_3, P_5, P_7$  — which are precisely the evidence that the true spacing is  $0.5\text{ Th}$ , not  $1.0\text{ Th}$  — contribute nothing to the  $z=1$  score and nothing against it. The matched filter is happy; the charge label is wrong. What is missing is any term that charges the  $z=1$  hypothesis for the half of the envelope it declines to explain.

## 2 The three requirements

A score fit for charge/monoisotope selection must combine three properties that the harmonic example shows to be individually insufficient:

- P1. Goodness of fit.** The observed intensities should match the *shape* of the theoretical isotope-ratio pattern, not merely be present.
- P2. Explained fraction.** Observed signal inside the window that the hypothesis does not predict should count against it; a hypothesis may not ignore half the peaks for free.
- P3. Completeness.** A predicted tooth that is absent from the data should count against the hypothesis; a half-present envelope is a worse explanation than a fully present one.

The central claim of this note is that these three requirements are not three terms to be weighted and summed. They are three behaviors of one quantity.

## 3 The metric

### 3.1 Union grid and formula

Let the *window* be a fixed  $m/z$  interval around the candidate monoisotope (Section 4). Define two index sets:

$$\begin{aligned}\mathcal{T} &= \{\text{predicted isotope positions of the hypothesis with weight above the significance cutoff}\}, \\ \mathcal{O} &= \{\text{observed peaks in the window above the noise floor}\}.\end{aligned}$$

Form the **union grid**  $\mathcal{G} = \mathcal{T} \cup \mathcal{O}$  by merging positions that coincide (within an  $m/z$  tolerance) and keeping the rest. On this common grid build two vectors indexed by  $j \in \mathcal{G}$ :

$$t_j = \begin{cases} \text{theoretical average weight at } j, & j \in \mathcal{T}, \\ 0, & j \notin \mathcal{T}, \end{cases} \quad o_j = \begin{cases} \text{observed intensity at } j, & j \in \mathcal{O}, \\ 0, & j \notin \mathcal{O}. \end{cases}$$

The score is the cosine similarity of these two vectors:

$$S = \frac{\sum_{j \in \mathcal{G}} t_j o_j}{\sqrt{\sum_{j \in \mathcal{G}} t_j^2} \sqrt{\sum_{j \in \mathcal{G}} o_j^2}} = \frac{\mathbf{t} \cdot \mathbf{o}}{\|\mathbf{t}\| \|\mathbf{o}\|} \quad (2)$$

with  $S \in [0, 1]$  for non-negative intensities.

### 3.2 How one formula delivers all three properties

The union grid is what makes Eq. (2) do three jobs at once. Every grid slot falls into one of three cases, and each case moves  $S$  in exactly the direction one of P1–P3 demands.

| Grid slot     |                                                             | $t_j$ | $o_j$ | Effect on $S$                                                                                                                  |
|---------------|-------------------------------------------------------------|-------|-------|--------------------------------------------------------------------------------------------------------------------------------|
| Matched tooth |                                                             | large | large | Raises the numerator $\mathbf{t} \cdot \mathbf{o}$ when the ratio $o_j/t_j$ agrees with the envelope shape ( <b>P1: fit</b> ). |
| Missing tooth | predicted ( $j \in \mathcal{T} \setminus \mathcal{O}$ )     | large | 0     | Contributes to $\ \mathbf{t}\ $ (denominator) but nothing to the numerator, so $S$ drops ( <b>P3: completeness</b> ).          |
| Unexplained   | observed peak ( $j \in \mathcal{O} \setminus \mathcal{T}$ ) | 0     | large | Contributes to $\ \mathbf{o}\ $ (denominator) but nothing to the numerator, so $S$ drops ( <b>P2: explained fraction</b> ).    |

Table 1: Each type of union-grid slot maps to one required property. There are no free parameters trading the three off; they are the same denominator/numerator bookkeeping.

Concretely, in the harmonic example the intervening peaks  $P_1, P_3, P_5, P_7$  enter the  $z=1$  score as ( $t_j = 0, o_j = I$ ) slots. They inflate  $\|\mathbf{o}\|$  with no numerator contribution and pull the  $z=1$  cosine down — exactly the penalty **RawSum** was missing.

### 3.3 Why $S^2$ is the explained-signal fraction

The “portion of signal explained” is not a bolted-on term; it is what Eq. (2) already computes. Model the observed vector as a scaled copy of the envelope plus residual,  $\mathbf{o} \approx A \mathbf{t}$ , and choose the amplitude  $A$  by least squares:

$$A^* = \arg \min_A \|\mathbf{o} - A \mathbf{t}\|^2 = \frac{\mathbf{t} \cdot \mathbf{o}}{\|\mathbf{t}\|^2}. \quad (3)$$

The energy of the observed signal splits into the part captured by the best-scaled envelope and the residual. Using  $A^*$ ,

$$\begin{aligned} \|\mathbf{o} - A^* \mathbf{t}\|^2 &= \|\mathbf{o}\|^2 - 2A^*(\mathbf{t} \cdot \mathbf{o}) + (A^*)^2 \|\mathbf{t}\|^2 \\ &= \|\mathbf{o}\|^2 - \frac{(\mathbf{t} \cdot \mathbf{o})^2}{\|\mathbf{t}\|^2}. \end{aligned} \quad (4)$$

Define the coefficient of determination through the origin,  $R^2 = 1 - \frac{\|\mathbf{o} - A^*\mathbf{t}\|^2}{\|\mathbf{o}\|^2}$  (the fraction of observed energy the scaled envelope removes). Substituting Eq. (4),

$$R^2 = \frac{(\mathbf{t} \cdot \mathbf{o})^2}{\|\mathbf{t}\|^2 \|\mathbf{o}\|^2} = S^2. \quad (5)$$

So  $S^2$  is *literally* the fraction of observed signal energy explained by the best-scaled average envelope, evaluated over a grid that already contains the unexplained peaks. Property P2 is therefore not an add-on: once the union grid includes the observed peaks the envelope ignores, the cosine’s square is the explained-energy fraction. ( $S$  and  $S^2$  are monotonically related, so the choice between reporting the cosine or its square is one of interpretation only.)

## 4 Design choices

Four judgment calls determine the metric’s behavior. They are where the real engineering lives.

**The window (what “should be explained” means).** The window sets the domain of  $\mathcal{O}$  — the set of observed peaks the hypothesis is held responsible for. We take it to span from the candidate monoisotope to  $\text{mono} + n \cdot \Delta$  (where  $\Delta$  is the isotope spacing and  $n$  the number of significant teeth) plus a small margin. **The same window, in  $m/z$ , is used for every charge hypothesis at a given monoisotope.** This is deliberate: a co-eluting interferent inside the window then penalizes every charge hypothesis equally, so it drags all absolute scores down together and leaves the *comparison* between charges fair. Using a per-charge window would let a hypothesis quietly shrink its own accountability. The window boundary is the most consequential parameter in the whole scheme.

**Noise threshold.** Before an observed peak is added as a ( $t = 0, o = I$ ) slot, it must clear a noise floor  $\eta$ . Without this, ordinary baseline noise in the window would read as “unexplained signal” and depress every cosine, muddying the comparison and biasing the metric against the true (fully-explaining) hypothesis. Peaks below  $\eta$  are simply not placed on the grid.

**Cosine, not Pearson.** Use cosine similarity, *not* the Pearson correlation. Pearson mean-centers each vector before the dot product; that subtraction destroys the meaning of a zero. An *absent* isotope peak should read as the number 0 (“no intensity”), but Pearson re-interprets it as “below the average intensity,” which is not the same thing and can even reward a hypothesis for a systematically empty tooth. The zero must mean zero for P2 and P3 to work, so mean-centering is disqualifying here. (FlashLFQ uses Pearson correlation for its binary accept/reject *gate*; that is a different task. For a fit-and-coverage *score* over a union grid with meaningful zeros, cosine is the correct choice.)

**Significance cutoff.** Which theoretical teeth count as “predicted” (the set  $\mathcal{T}$ ) is set by a weight cutoff — e.g. keep teeth whose average weight is  $\geq 0.2$  of the modal tooth. Teeth below the cutoff are negligible in the theoretical pattern; including them would add near-zero  $t_j$  slots that only add noise to the completeness accounting.

## 5 Validation

We evaluated the metric on peptide **EFLDVIK**, true charge 2, monoisotopic mass 862.488 Da, with 15 observed peaks inside the window. The existing detector had *mislabelled* this feature as charge 1 at mass 431 Da — the harmonic error of Section 1. Table 2 reports, for each charge hypothesis, completeness (fraction of predicted teeth present), the explained fraction (%), and the union-grid cosine  $S$ .

| Hypothesis                       | Mass (Da) | Completeness | % explained | cosine $S$ (union) |
|----------------------------------|-----------|--------------|-------------|--------------------|
| $z = 1$ (harmonic)               | 431       | 1.00         | 0.60        | 0.905              |
| <b><math>z = 2</math> (true)</b> | 862       | 1.00         | 0.76        | <b>0.984</b>       |
| $z = 3$ (over-split)             | 1294      | 0.33         | 0.54        | 0.718              |

Table 2: EFLDVIK charge selection. The unified cosine ranks the true charge first; neither sub-metric does so alone.

**What the table shows.** Read column by column, the sub-metrics each fail on their own:

- **Completeness alone ties  $z=1$  and  $z=2$**  (1.00 vs 1.00). Both the true charge and its harmonic have all their predicted teeth present — of course they do, since the  $z=1$  teeth are a subset of the  $z=2$  peaks. Completeness cannot break the harmonic tie and cannot fix the mislabel. It does, correctly, sink  $z=3$  (0.33), which predicts teeth at 0.33 Th spacings that are largely absent.
- **Explained fraction alone catches  $z=1$**  ( $0.60 < 0.76$ ), because the harmonic ignores the intervening peaks — but on its own it does not cleanly separate the field ( $z=3$  at 0.54 is close to  $z=1$  at 0.60, for the unrelated reason that  $z=3$  predicts too much).
- **The unified cosine separates all three.**  $z=1$  is pulled down to 0.905 by the unexplained intervening peaks (P2);  $z=3$  is pulled down to 0.718 by its missing teeth (P3);  $z=2$  retains a high 0.984 because it both explains the observed peaks and has all its predicted teeth present (P1–P3 together). The true charge wins.

This is the argument for a single cosine-over-union score rather than any one sub-metric or an ad hoc weighted blend: the three requirements are already co-resident in Eq. (2), and only their combination ranks the hypotheses correctly.

## 6 Caveats and implementation note

**Caveats (stated honestly).**

- **The winning margin is modest.** 0.984 vs 0.905 is a real separation but not a chasm. Reliable charge *re-selection* across a full dataset needs empirical validation before it replaces the existing decision: on genuinely noisy or sparse envelopes the metric could begin to prefer higher charges (which predict more, finer teeth and can accrue spurious matched intensity). This must be measured, not assumed.
- **The window boundary dominates.** As noted in Section 4, the choice of window is the single most consequential parameter; it defines what counts as “unexplained” and therefore the entire P2 penalty. It deserves the most tuning and the most scrutiny.

- **Chimeric envelopes lower the cosine — correctly.** When two features co-elute and overlap, no single-envelope hypothesis explains all the in-window signal, and every cosine is depressed. This is the *right* signal: a uniformly low  $S$  across all charges is a flag to route the window to advanced multi-envelope deconvolution rather than to force a single-charge label.

**Implementation note.** The detector already contains most of this. The opt-in scorer `ScoreModel::Normalized` with the `score_cosine` flag computes a cosine, but currently only over the *comb* slots — i.e. over  $\mathcal{T}$  alone. Over  $\mathcal{T}$  the cosine already delivers P1 (fit) and P3 (completeness, via the missing-tooth ( $t>0, o=0$ ) slots), but it is blind to P2 because unexplained observed peaks off the comb are never on the grid. The single change that upgrades it to the full three-property metric is to add the in-window observed peaks that are *not* on the comb as ( $t = 0, o = I$ ) slots (after the noise-floor filter) before taking the cosine. That one addition to the grid turns the existing comb cosine into Eq. (2), and with it  $S^2$  becomes the explained-energy fraction of Section 3.